

Name Redaction Does Not Blind an Arabic Résumé Ranker: An Open, Pre-Registered Instrument and Power Analysis for Gender-Bias Audits in AI-Assisted Recruitment

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1 NILES 2026 - SS2 Submission Proposal

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1.1 Introduction

Every published fairness safeguard for automated résumé screening assumes that removing the applicant’s name removes the protected attribute. Arabic breaks that assumption at the level of grammar: agreement on verbs, adjectives and participles marks the applicant’s gender obligatorily and repeatedly inside the CV (تخرجت / تخرج, حاصلة / حاصل), where no employer edit and no redaction reaches. This paper does not claim a bias effect size. It delivers the three things that must exist before any such claim can be believed — a measurement of how pervasive the marking actually is in real Arabic hiring text, an open and cryptographically frozen audit instrument, and a power analysis showing that the equivalence claims routinely made in counterfactual hiring audits are unattainable at feasible sample sizes — and it registers the confirmatory study publicly, before unblinding.

1.2 Contributions

C1 — Cue-level measurement of obligatory gender marking in real Arabic job ads. A released morphological tagger identifies gendered referring expressions and, critically, decides whether each one refers to *the applicant* — خبرة واسعة (”broad experience”) is grammatically feminine but describes the experience, not the person. The tagger abstains rather than guesses on genuine Arabic ambiguity and routes those cases to human adjudication. Applied to the ArabJobs corpus (CC-BY; Egypt, Jordan, Saudi Arabia, UAE), it yields the first cue-level prevalence statistics for applicant-referring gender marking in Arabic recruitment text, by country, seniority and occupation, with tagger precision/recall on an adjudicated sample. This establishes empirically — not rhetorically — that the redaction safeguard is structurally inapplicable to Arabic.

C2 — ArabGN-CF: an open, pre-registered audit instrument. Approximately 7,900 lines, 117 automated tests, executable end-to-end on a laptop with no model downloads via a deterministic synthetic scoring backend, so a reviewer can re-run every estimator against known ground truth. Components: a five-register ad typology (as-found generic masculine · dual/inclusive · agreement-free · syntax-matched masculine placebo · cross-lingual English) with machine-checked register invariants; a gender-controlled twin CV generator that *refuses* to emit a pair differing in token count; SD-matched competitor pools; HMAC-based blinding of cell identity from everyone who prepares or scores material; a cryptographic freeze over the config *and every analysis module’s source*, after which confirmatory analysis refuses to run on drift; and a reporting layer that refuses to emit the phrase ”no bias”, because no test can license it.

C3 — A power and feasibility result for counterfactual hiring audits. For a twin contrast over A ads $\times C$ CV twins, $SE^2 = \sigma_{ad}^2/A + \sigma_{cv}^2/C + \sigma_{resid}^2/(A \cdot C)$. Three consequences, each transferable beyond Arabic: (i) the ad axis — the expensive, human-authored one — *caps* attainable precision, so buying more ads cannot reach a tight equivalence margin at any number of CVs; (ii) at a top-10-of-100 selection base rate the principled margin $\min(2 \text{ pp}, 0.10 \times \text{base}) = 1 \text{ pp}$ is **not powerable** on the selection-rate outcome at feasible cost (≥ 854 ads ≈ 85 h of expert authoring), so only the four-fifths adverse-impact benchmark is reachable, and reaching it requires **770 twins**, not more ads; (iii) treating competitor-pool resample draws as units of analysis — a common practice — understates the standard error by more than an order of magnitude, whereas two-way cluster-robust variance over ads and twins recovers a $1.6\times$ inflation on real structure. We report the variance components measured in a 40-ad $\times$ 50-twin pilot rather than assumed.

C4 — The pre-registration itself, published before unblinding. Cells, margins and their derivation, the Holm family, the positive control, the zero-cue calibration cell, the pre-committed lexical-deflation test (BM25 over the ad registers), and the decision rule that *declines* localisation if score-level and selection-level outcomes order the cells differently. Publishing this at NILES timestamps the commitment, with the freeze hash printed in the paper. Pre-registration is common in the social sciences and rare in machine-learning audits; making it inspectable is part of the contribution.

1.3 Audit subjects and scope

Subjects are frozen, version-pinned, open-weight systems that we measure and never train: multilingual-e5-large, BGE-M3, jina-v3, an Arabic encoder, and two open-weight LLM screeners. Nothing we build is an audit subject. Scope is Modern Standard Arabic; dialectal CVs are out of scope. Gender is treated as binary because Arabic agreement is, and the agreement-free register is the closest the design comes to a neutral form.

1.4 What this paper deliberately does not report

The confirmatory twin contrasts remain sealed. The pilot is pre-specified to estimate dispersion and twin-discordance only, and is forbidden from touching any hypothesis, margin or decision rule; we therefore report variance components and withhold signed effects. Stating this plainly is a strength, not a gap: disclosing a first-look effect would void the design the paper is registering. We also make no claim about any commercial hiring system, and none about real-CV population statistics — the twins are synthetic by necessity, since no two real applicants have identical qualifications.

1.5 Fit to SS2

The session’s three terms map directly onto the paper. *Fairness and bias*: a protected-attribute channel that the field’s standard mitigation cannot close, in a language with $\sim 400\text{M}$ speakers. *Responsible LLMs in recruitment*: open-weight LLM screeners audited under blinding and a frozen analysis, with a reporting layer that refuses overstatement. *Policy*: regulators treat hiring AI as high-risk and treat redaction as a safeguard; this tests whether the safeguard functions in Arabic and quantifies the cost of the alternatives, against the EEOC four-fifths benchmark (29 CFR 1607.4(D)). Venue relevance is direct: the human-side remedy this work tests against a machine is Soliman & Sczesny (2026, *EJSP* 56(3):507–520), $N = 181$ **Egyptian** recruiters.

1.6 Positioning against prior art

ArabJobs (arXiv:2509.22589) and JobArabi (arXiv:2605.20960) audit Arabic **ads**; we audit the **ranker**, and use ArabJobs as a dependency, claiming no Arabic-resource first. García-Sardiña et al. (arXiv:2509.13803) study grammatical-gender bias in retrieval for Spanish, French, German and Italian — **Arabic absent**, unit = job **titles**, not documents. Goyal et al. (arXiv:2604.06097) establish that query rewriting can reduce retriever bias *through increased score variance*, plus an interaction-vs-encoding taxonomy; we cite it and adopt both as **method** (SD-matched pools, per-cell variance reported), on English format biases with no protected attribute and no morphology. Arabic gender rewriting (APGC v2.0, arXiv:2110.09216; Alhafni et

al., arXiv:2205.02211) is sentence-level 1st/2nd-person $M \leftrightarrow F$ with **no neutral target**; the agreement-free register and its tagger occupy that empty lane.

1.7 Feasibility, 3 → 20 August 2026

Dates	Work	Status
3–5 Aug	Confirm SS2 format, page limit and template; send ArabJobs redistribution query	Not started
4–7 Aug	Phase 1 pilot (40 ads × 50 twins, e5-large; GPU-minutes, zero human hours) → measured variance components, re-derived N	Instrument ready, blocking
6–10 Aug	Tagger sweep over ArabJobs; adjudicate an abstention sample (≈4 h in-team) → C1 tables	Tagger implemented
8–14 Aug	Draft	—
15–17 Aug	arabgn freeze ; hash into the manuscript; internal review	—
18–19 Aug	Submit, one-day buffer	—

Phase 0 of the instrument is complete and tested. No component of this paper requires the 30 h ad-authoring budget, human relevance annotation, an ethics approval, a recruiter panel or a paid API — every one of those sits on the confirmatory study’s critical path, not this one.

1.8 Risks and mitigations

1. **”Protocol paper without findings.”** C1 and C3 are findings, both checkable and both independent of any bias effect existing. The paper is framed as measurement and feasibility, not as a protocol stub.
2. **A reviewer asks for the audit result.** Answered in the design: the result is sealed under a published pre-registration, and the freeze hash lets a reviewer verify that the analysis predates unblinding.
3. **ArabJobs redistribution unconfirmed.** C1 reports aggregate statistics only; release of derived ad variants is deferred to the journal paper. The licence question is therefore off this submission’s critical path.
4. **Page limit.** The power derivation and register invariants compress into a table plus a repository pointer; the arithmetic is code and is asserted by the test suite.